



Universität St.Gallen

Part 2 Course Project: KAFKEA Restocking

EDPO - Event-Driven and Process-Oriented Architectures

St.Gallen, 22.05.2026

Eva Amromin, Gianluca Ielpo, Michael Schütz



Source image: <https://chatgpt.com/>

From insight to impact.



Project Overview



System Architecture (Topologies)



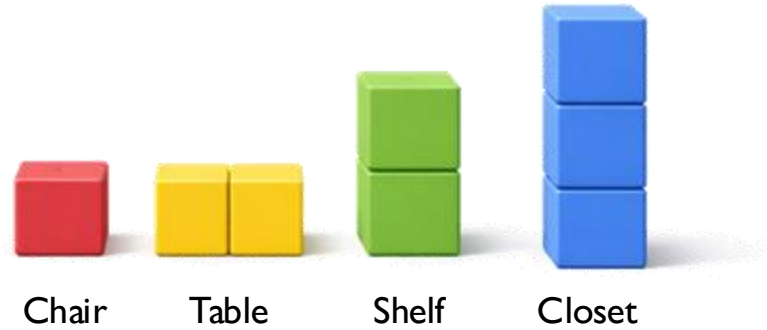
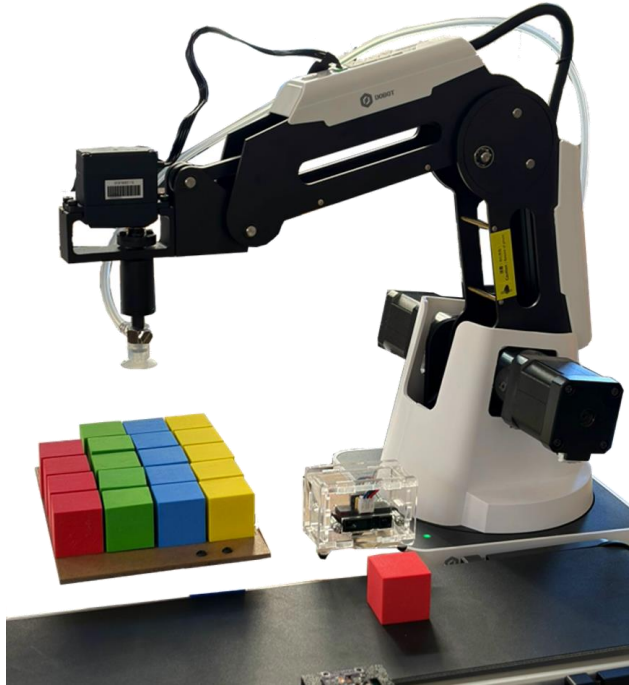
Demo



Future Integration & Takeaways

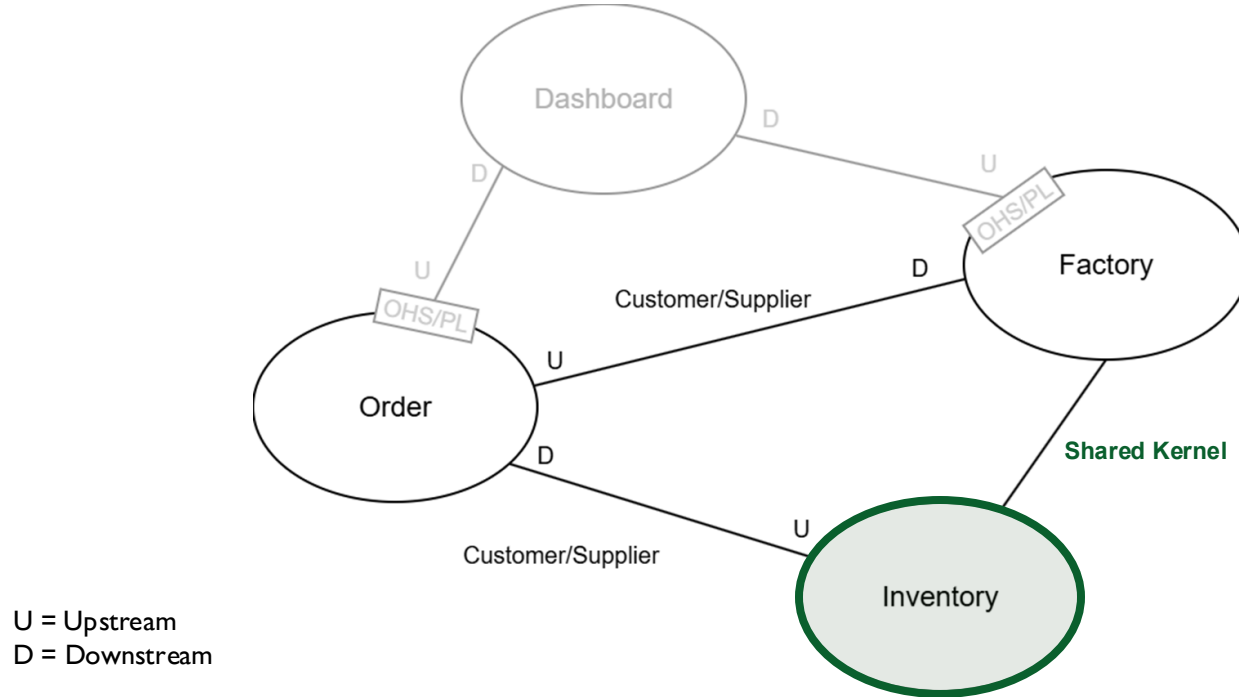
Agenda

Recap



Source image: <https://chatgpt.com/>

System Overview



Our next goal: implement the restocking of the inventory system using event streaming

What's new in Assignment 1

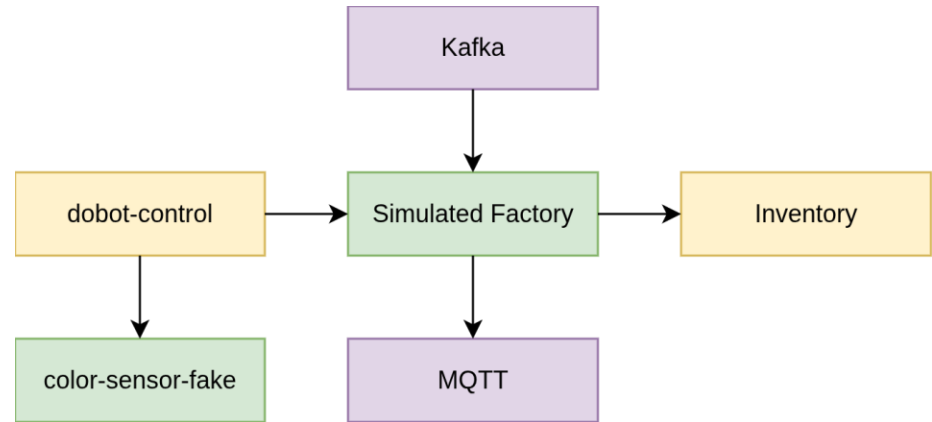
Changes from Assignment 1 – KAFKEA Simulator

- No implementation changes to KAFKEA
- Implemented a more complete KAFKEA simulator
- Minor additions to documentation (incl. one ADR)

Source image: <https://chatgpt.com/>

Changes from Assignment 1 – KAFKEA Simulator

- Goal: allow the development and testing of the software without requiring access to hardware
- Reproduce behavior
- Controlled error injection
- Visualization of state and events
- Run preconfigured flows



Source image: <https://chatgpt.com/>

Presets

Run a deterministic scenario and watch state and events update in real time.

happy-path

preset

Deterministic successful order flow with matching color and IR detection.

Run happy-path

wrong-color

preset

The flow continues, but the color sensor deterministically returns blue.

Run wrong-color

pickup-failure

preset

The IR sensor never detects an item, reproducing a failed pickup.

Run pickup-failure

Stop Reset

Factory Twin

Live block diagram of the simulated factory. Active step: —

Inventory Grid

read-only · from
inventory service

Unavailable

Conveyor Belt

Direction STOP
Speed 0.0
Distance 0.0

Robot (left)

read-only

Position x 0.0
Position y 0.0
Position z 0.0
Position r 0.0
Speed 50.0
Suction off

Color Sensor

color-left fixed

Mode
fixed

Color
RED

Raw RGB
1
0
0

Apply

Distance / IR Sensors

distance-
conveyor scripted value:
30.0

Mode
scripted

Fixed value
30.0

Scripted values
30.0

—
12.5

—
6.2

—
30.0

Add

Apply

ir-left fixed value: true

Assembly Area

Active step —
Step # 0

Assignment 2

Course Concepts Mapped to Our Implementation



KStream

Distance readings and color readings each arrive as continuous KStreams



KTable – Stateful Operation

Used to store the inventory and to perform edge detection



Interactive Queries

GET /inventory reads the inventory-store KTable directly



Custom Windowing

Session windows do not fulfill our requirements, implemented using Processor API



Stateless Operations

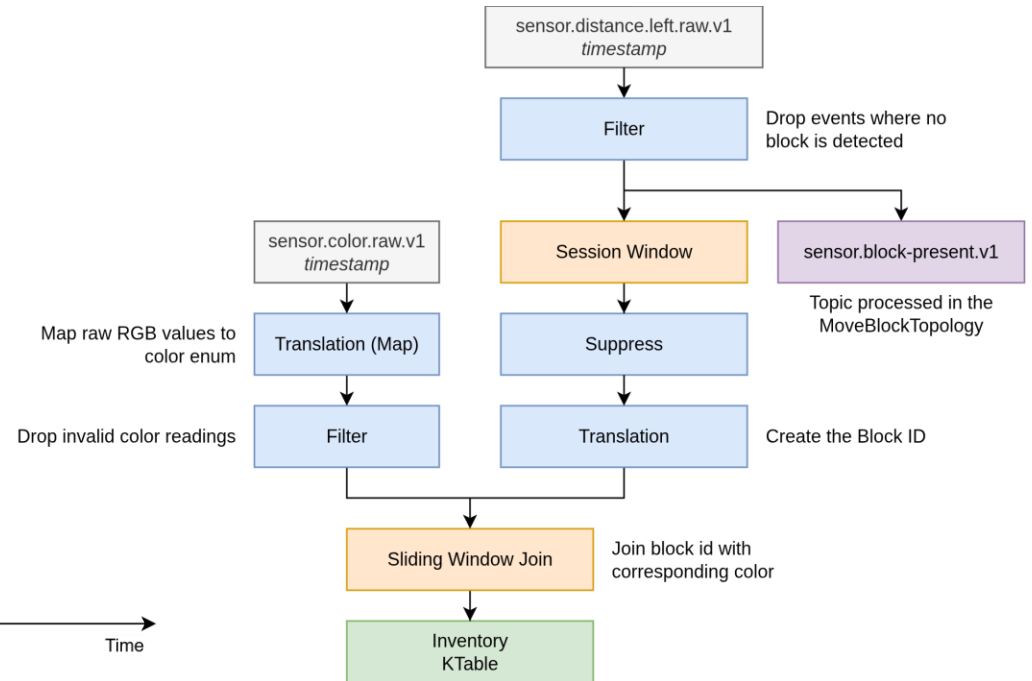
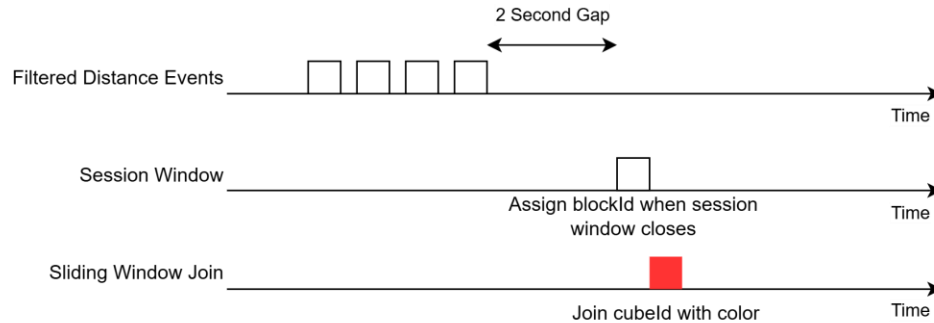
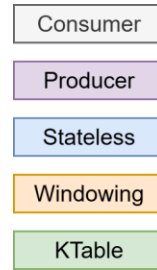
Filtering of color values, mapping of RGB values to enum, mapping of distance readings to true/false



Choreography

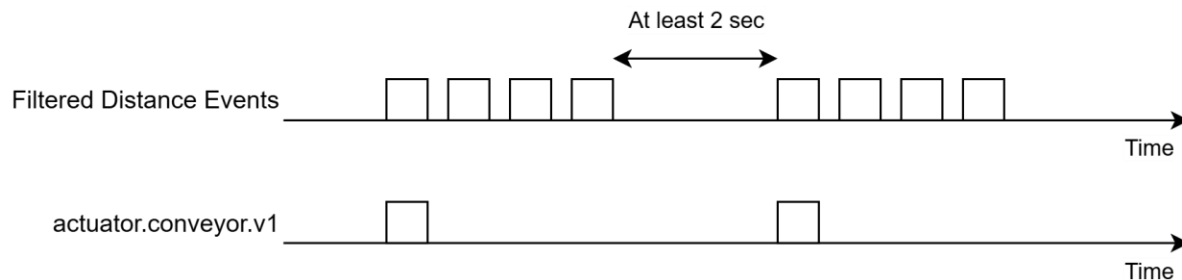
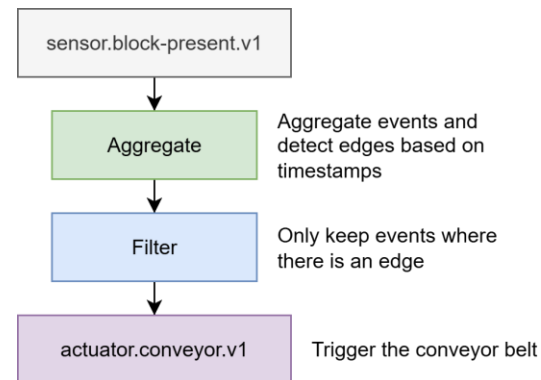
Placing the block in the inventory is achieved through choreography of multiple services

Kafka Streams Topology Diagram 1



Kafka Streams

Topology Diagram 2



Integration

How would we integrate the restocking into our existing system?

- Inventory grid no longer position-ordered, blocks fill first available slot

Process: Toggling topology

- On every inventory item restock/removal:
Send command to start/stop publishing distance events

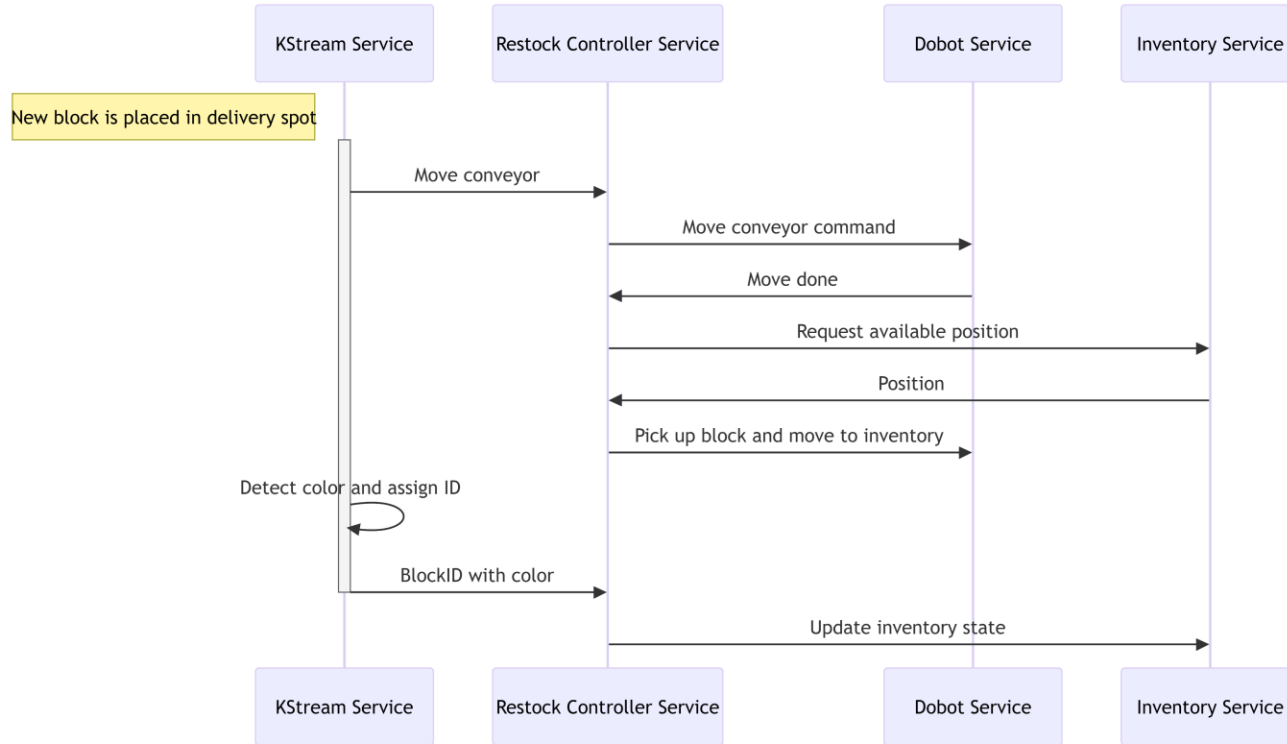
Process: Adding blocks to inventory

1. New endpoint to get next available position
`GET /restock/available-position`
2. Dobot moves block to inventory slot
3. Updated endpoint ensures inventory state
`POST /restock`



Source image: <https://chatgpt.com/>

Restocking Sequence



Demo

Key takeaways



Creating topology seems simple, but the whole scaffolding is more involved than expected.



There are many solutions to one problem, and the trade-offs are central in choosing.



Finding a problem to solve with a given tool can be challenging.



Building simulator is more involved than foreseen.



Kafka windows do not emit event directly when time expires, needed to use Processor API.

Thank you. Questions?



School of Computer Science (SCS-
HSG)
Rosenbergstrasse 30
9000 St. Gallen

unisg.ch/de/universitaet/schools/informatik